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(COA)

BTech CSE (DS)

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ASSIGNMENT - 4

Concept explanation type

Ans1 SRAM Vs DRAM

- SRAM (Static RAM): Faster, no ~~replies~~ refresh needed, expensive, used in Cache
- DRAM (Dynamic RAM): Slower, needs periodic refresh cheaper, used in main memory.

Ans2 Locality of reference: It means a program tends to access the same data or nearby data repeatedly within a short time.

- Temporal locality: Same data used again soon.
- Spatial locality: nearby data accessed soon.

Ans3 Cache Memory:
used to store frequently accessed data to reduce access time and improve CPU performances.

Ans4 Cache Indexing Mechanism:

- Direct Mapping: One memory block → One cache line.
- Fully Associative: → any block → any cache line
- Set Associative: block → any line within a set.

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Ans 6 ISA is the interface b/w hardware and software defining supported instruction.

Include
- Instruction format
- Data type
- Registers and addressing mode

Ans 7 Type of instruction in a processor are-
- Arithmatic (ADD, SUB)
- Logical (AND, OR)
- Data transfer (LOAD, STORE)
- Control (BRANCH, JUMP)

Ans 8 Explanation
- Opcode is interpreted
- Required operation is identified
- Control signal are generated
- Operand are prepared for execution

Ans 9 Basic instruction cycle are :-
- Fetch instruction from memory
- Decode instruction
- Execute operation
- store result

Ans 10 Characterstic of RISC processor are
- Simple and fixed - length instruction
- few addressing mode
- Easter execution (pipeline friendly)
- Load / store architecture

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Ans 11 Characterstic of RISC processor are,
- Simple and fixed - length instruction
- few addressing mode
- Easter execution (pipline friendly)
- Load / store architecture.

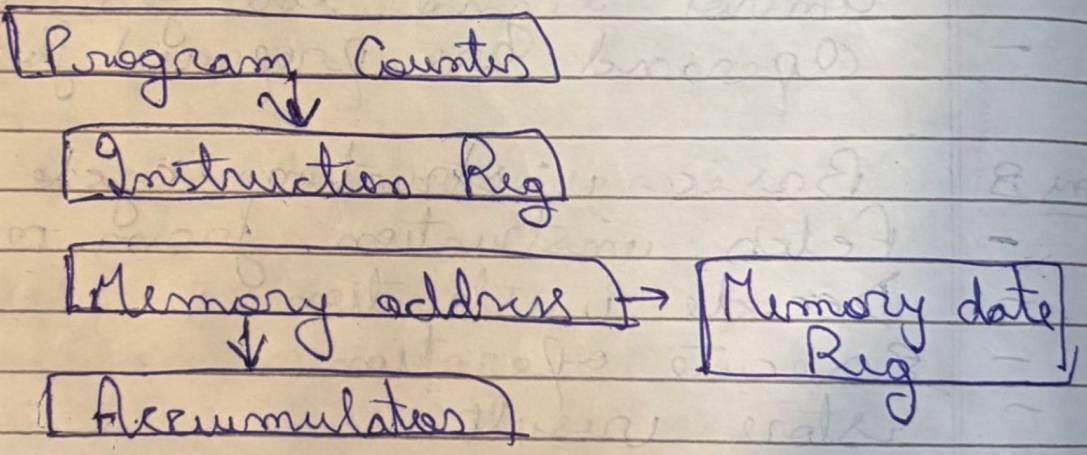
Ans 12 Risc - V

- Open - source ISA.
- Simple, fixed length instruction
- Easter to design and extend

X 86
- Proprietary (Intel / AMD)
- Complex, variable - length instruction

|| SECTION - B ||

Ans 1



- PC holds next instruction address, IR store current instruction.
- MAR & MDR handle memory access
Acc store intermediate result.

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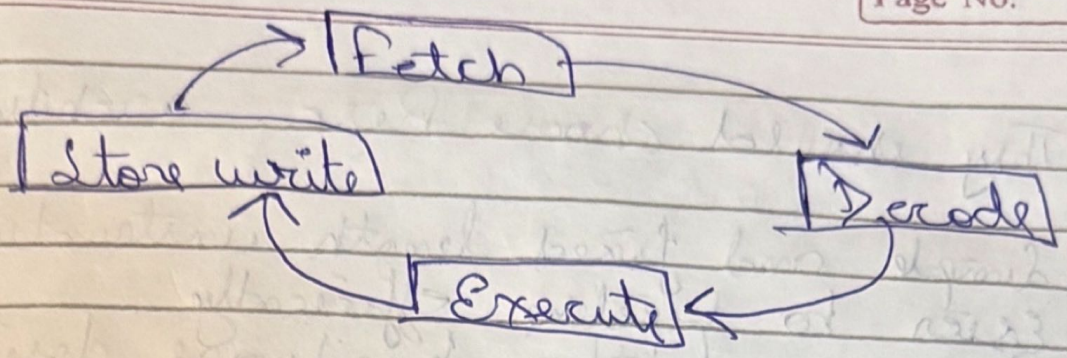
Ans 2



- CPU fetches instruction, decode it, execute it, and write Back.

Ans 3 Opcode / operands

- Opcode specify the opperation
- Operand specify data or memory location

Ans 4

Feature	Risc	Cisc
Instruction	Simple	Complex
Length	fixed	variable
speed	fast	slower
memory	more usage	less use

- Risc focus on simplicity and speed
- Cisc focus on powerful and complexity

SECTION - C

Ans 1 Impact on perfomance are

- Memory access is slower than registers access
- Increased CPI due to memory latency
- Overall execution time increase = performance decrease.

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Ans2 They should choose Risc architecture bec

- Simple and fixed length instruction
- Easier to pipeline efficiently
- Requires simpler hardware design.

Ans3 CPU executes this instruction because

Step 1 Fetch
Instruction is fetch from memory to I

Step 2 Decode
opcode is decoded and register are identified.

Step 3 Execute
Data is read from register
ALU perform operation
Result is stored back in register.

Ans 4 X 86 is widely used despite complex because :-

- Large ecosystem (OS, application, industry support)
- Modern CPU's internally optimised
- Despite complexity, compatibility and ecosystem keep X 86 dominand.

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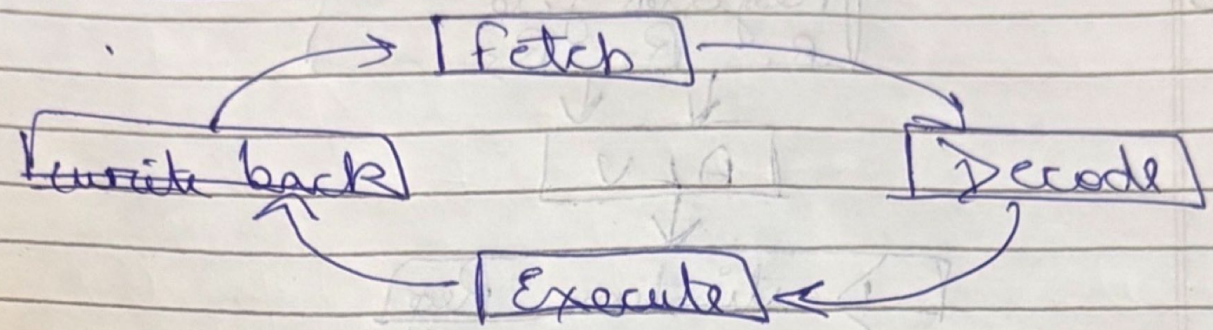
SECTION - D

Ans 1



Fetch
- PC holds address of next instruction
- Address sent to memory via MAR
- Instruction fetch into MAR and loaded into IR
- PC is incremented to meet instruction

Ans 2 funct 7 rs2 rs1 funct 3 rd opcode

- opcode - specific operation type
- rd - Destination register
- rs1, rs2 - Source register
- funct 3, funct 7 - Define exact operation
- fixed length [32-bit] format simple decoding

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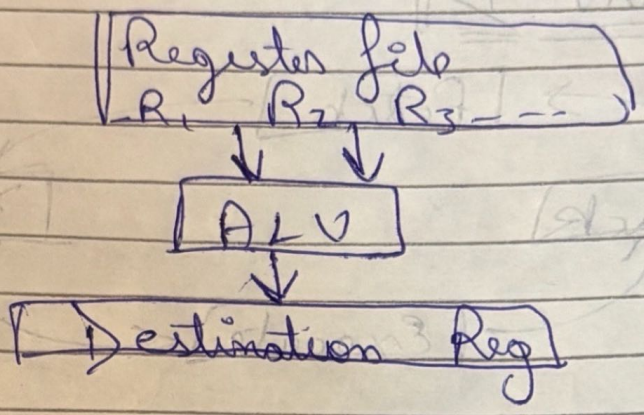
Ans 3



- Data is red from source registers
- Sent to ALU for operation
- Result is written back to destination reg
- Enable fast computation without memory access.